

Basic Principles of Medicine 2

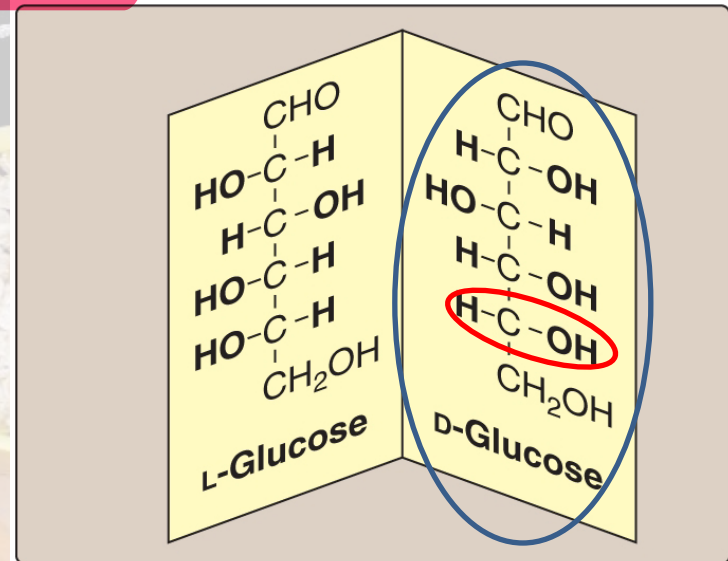
Module : Digestion and Metabolism

DLA Title: Structure and function - Carbohydrates

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Overview of functions of carbohydrates

- Energy in diet (45-65%)
- Storage form: Glycogen in liver and muscle
- Dietary fiber: Cellulose
- Glucose in blood (Brain; RBCs, WBCs, tissues of eye)
- Connective tissue/ cell membrane
- D-forms of sugars

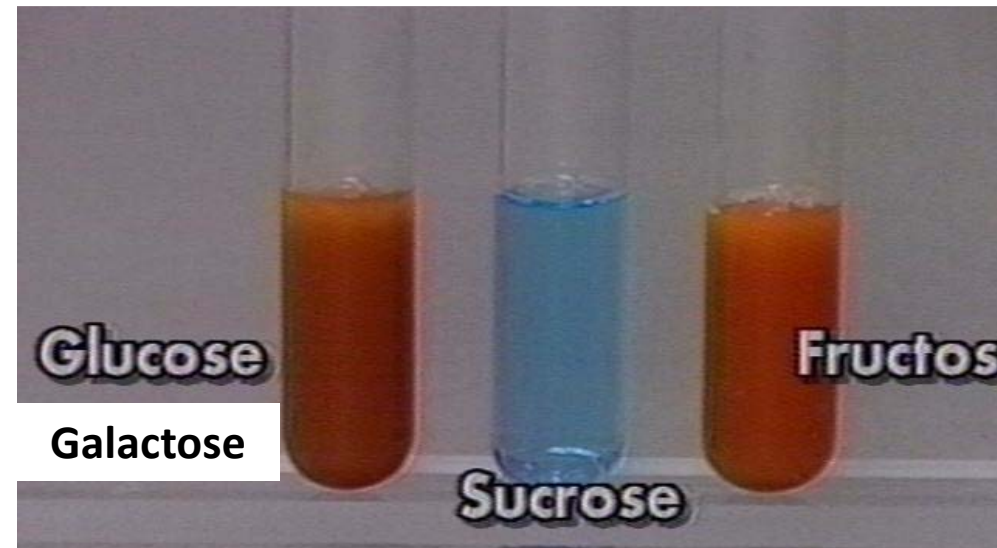
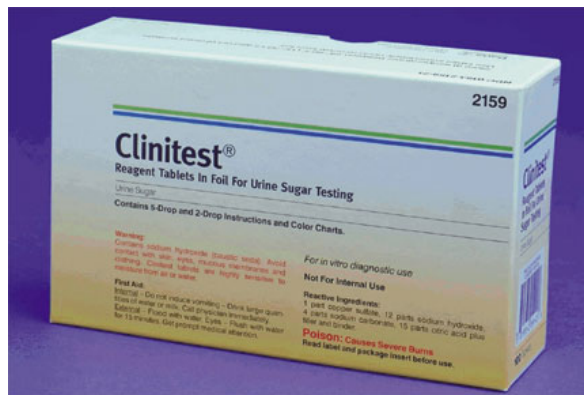
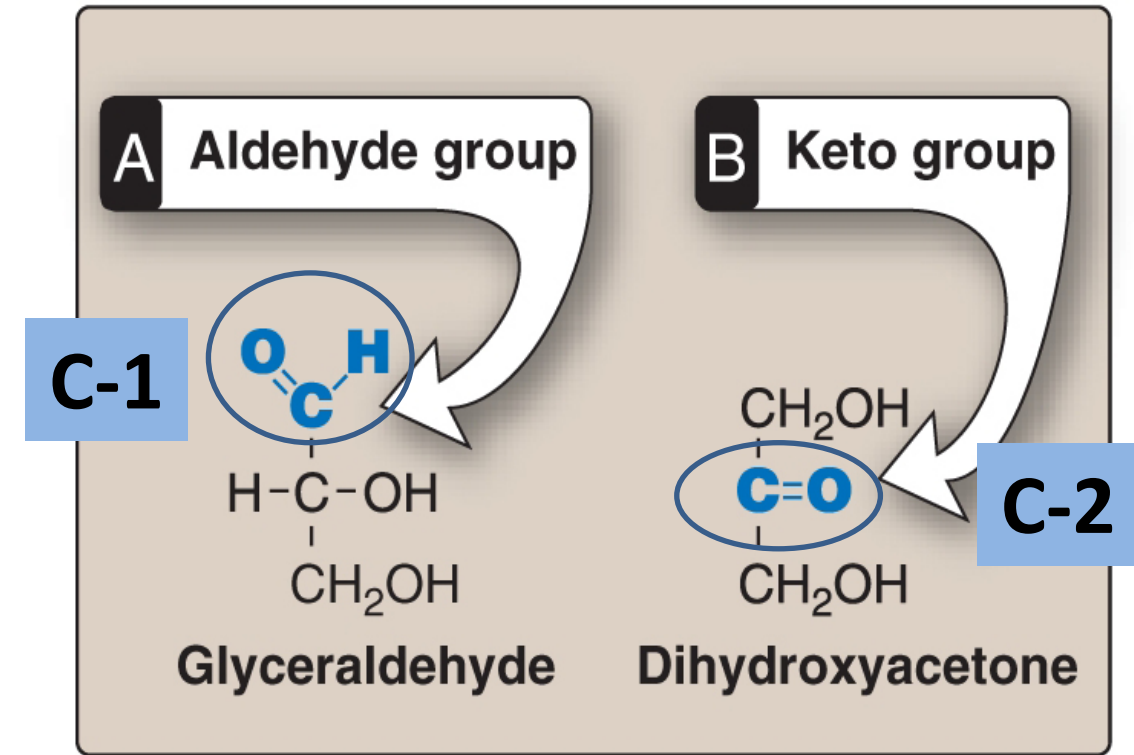


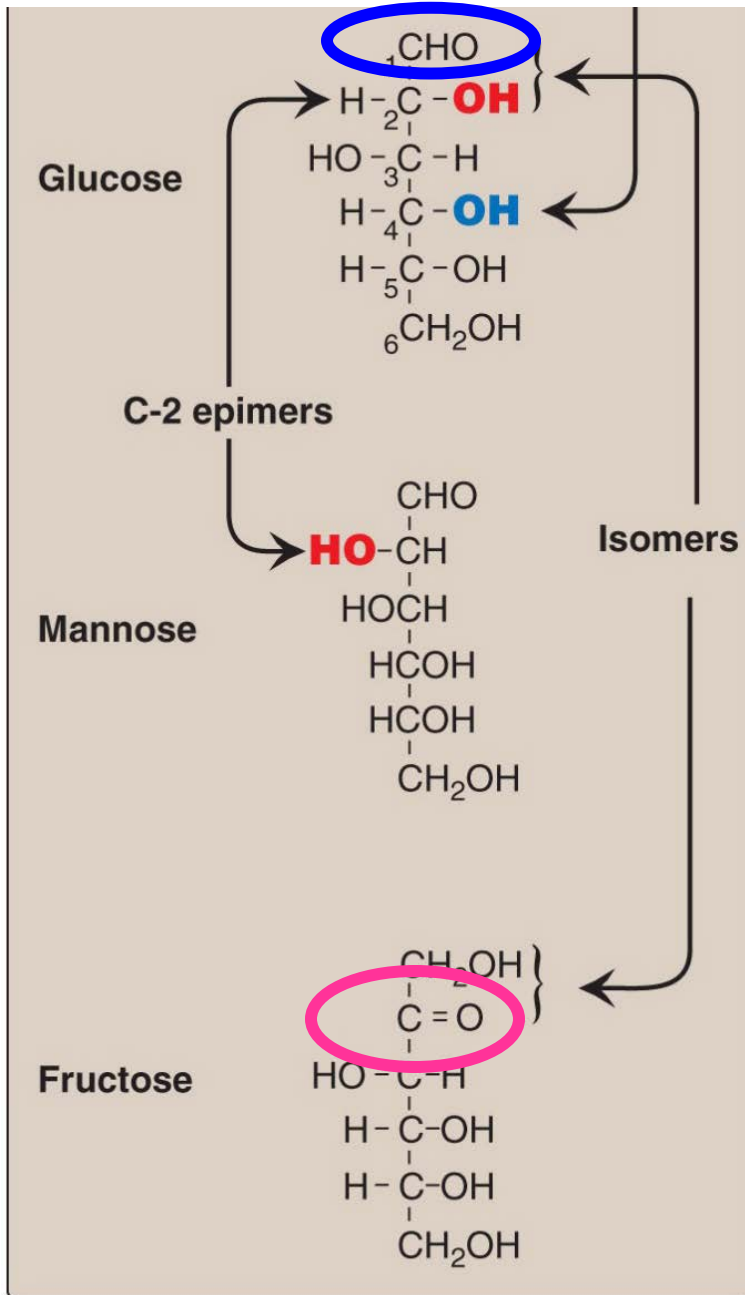
Monosaccharides: Functional group and C-atoms

	Aldose = Aldehyde @ C1	Ketose = Ketone @ C2
3 carbon (Triose)	Glyceraldehyde	Dihydroxyacetone
5 carbon (Pentose)	Ribose Xylose	Ribulose Xylulose
6 carbon (Hexose)	Glucose Galactose Mannose	Fructose

Reducing sugars

- Free aldehyde or keto group
- Reducing sugars in urine
 - Convert cupric ions to cuprous ions
- Aldoses vs Ketoses





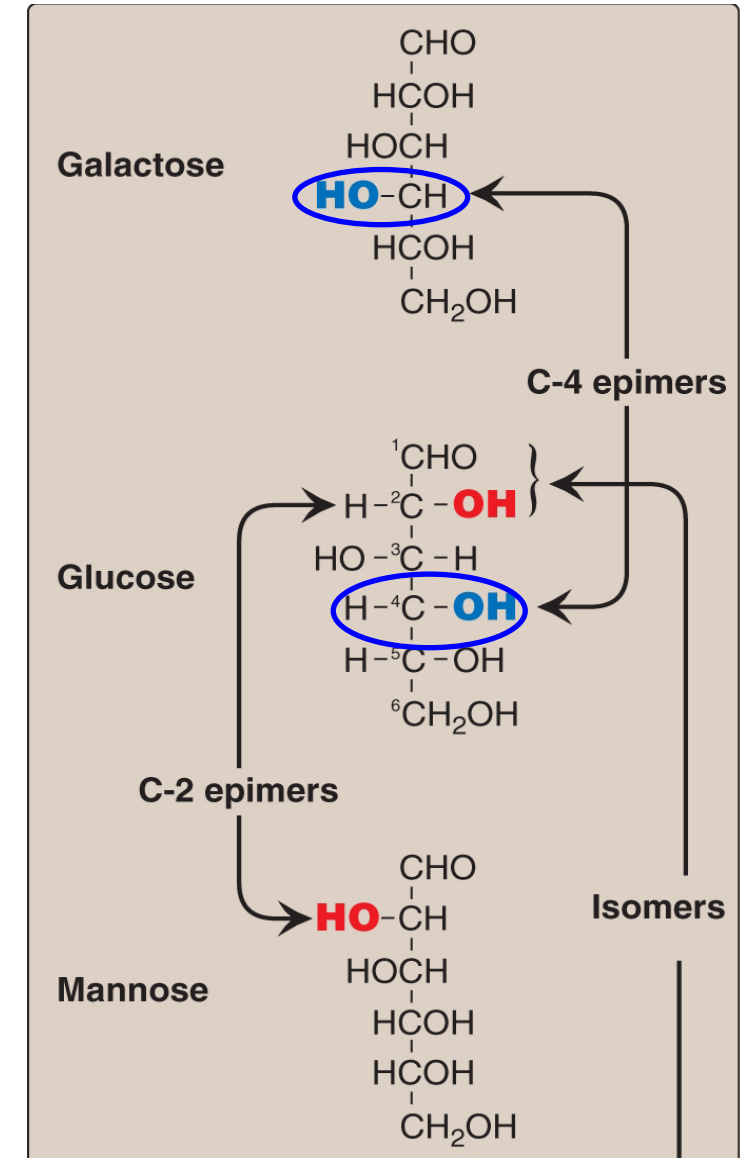
Glucose vs Fructose

(Aldohexose vs Ketohexose)

- Phosphohexose isomerase (glycolysis)
- Reducing sugars

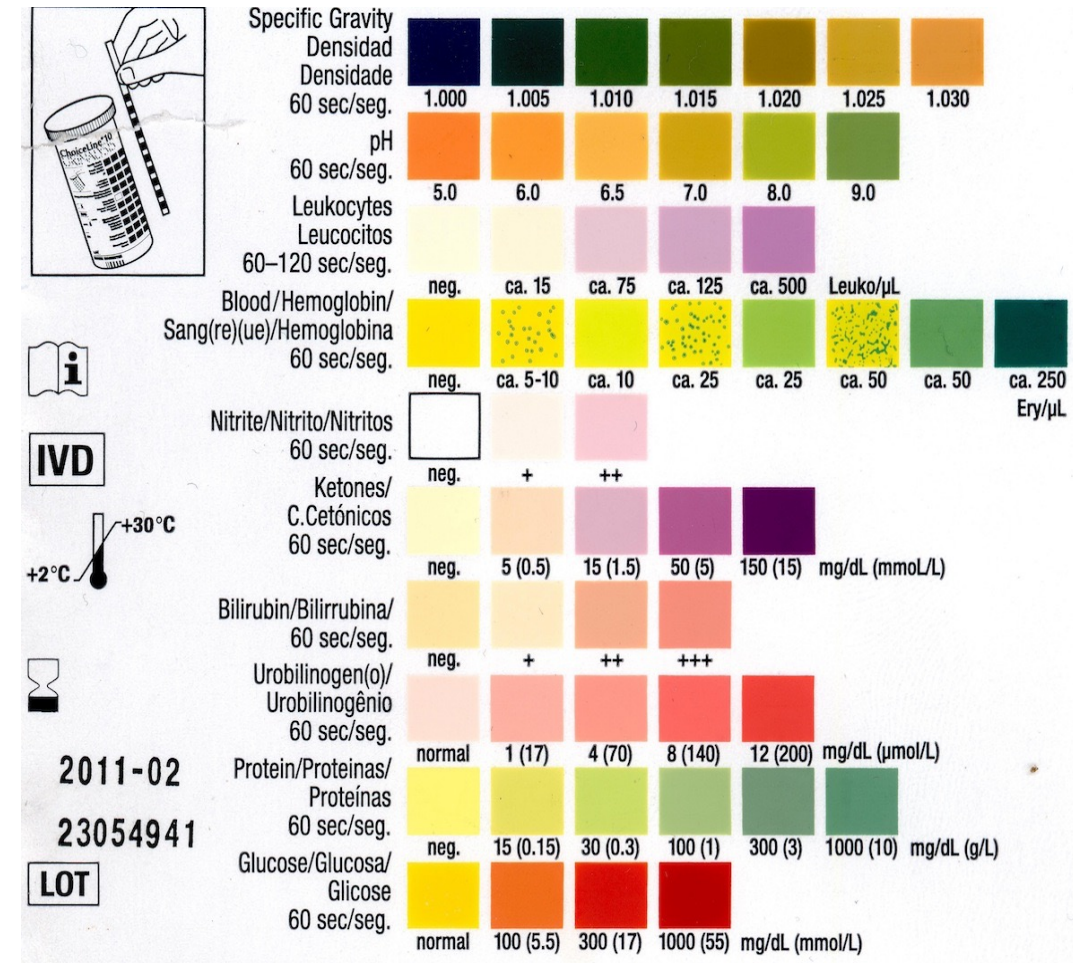
Galactose

- C-4 epimer of glucose
- Aldohexose
- Epimerases (galactose metabolism)
- Avoid galactose: Galactosemia



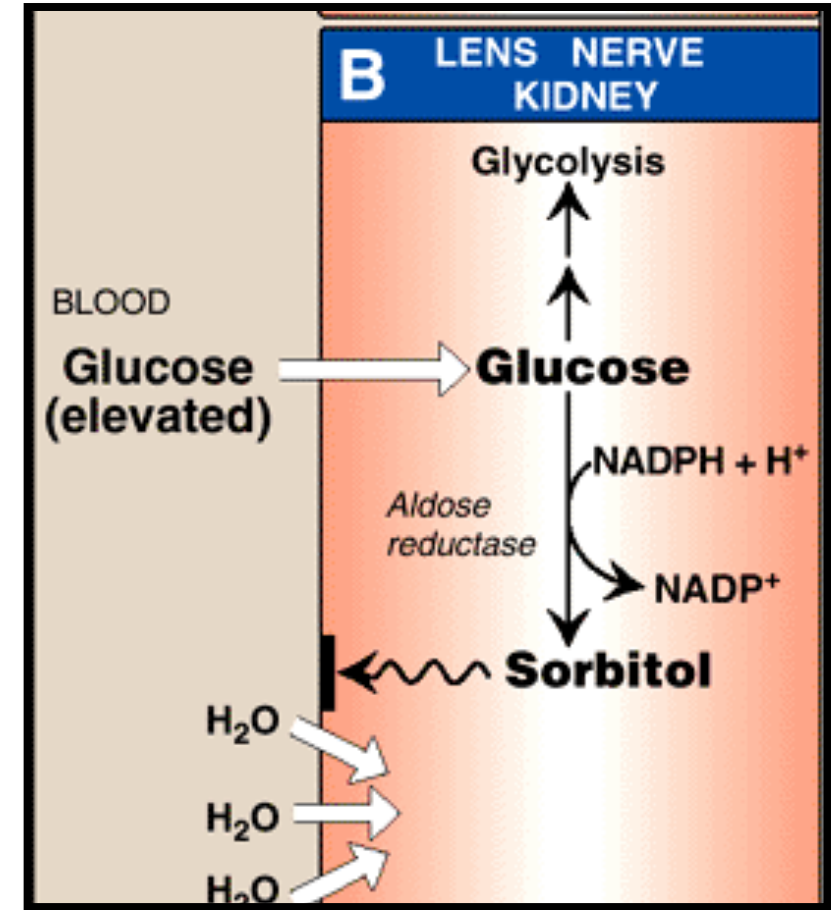
Sugars in urine: Clinical significance

- Sugars absent in urine of a normal person
- Reducing property
 - Benedicts test or Clinitest
- Sugars in urine:
 - Glucose: Diabetes mellitus
 - Fructose
 - Galactose
 - Lactose intolerance (no sugar in urine)



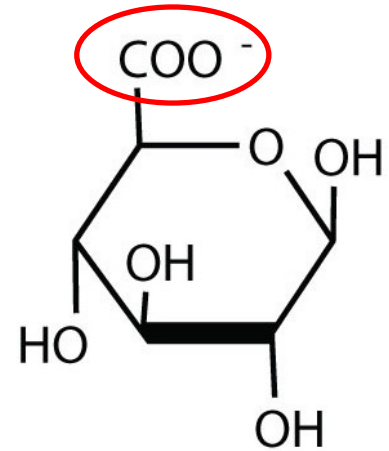
Polyols (sugar alcohols)

- **Sorbitol**: Sugar alcohol from glucose
 - Diabetes Mellitus
- **Galactitol** formed from galactose
 - Galactosemia
- **Inositol** present in IP3 (second messenger)



Glucuronic acid

- Oxidation of C-6 of glucose
- Negatively charged
- UDP-glucuronic acid: Conjugation reactions (bilirubin)



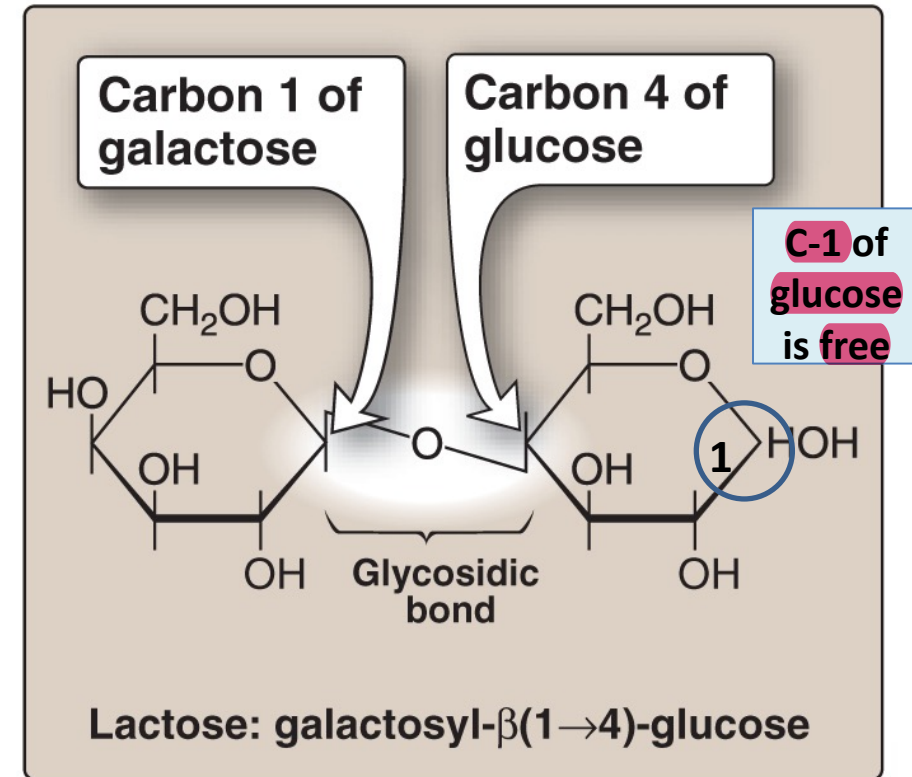
Glucuronic acid

Classification of carbohydrates

- **Monosaccharides: Monomers**
- Based on number of monomers:
 - Monosaccharides
 - **Disaccharides**
 - Oligosaccharides
 - Polysaccharides

Lactose

- Galactose + ^{C1-free} Glucose
- $\beta 1 \rightarrow 4$ glycosidic linkage
- Reducing sugar
- Intestinal lactase (brush border disaccharidase)
- NOT found in urine in lactose intolerance
- Avoid Lactose: Lactose intolerance and galactosemia

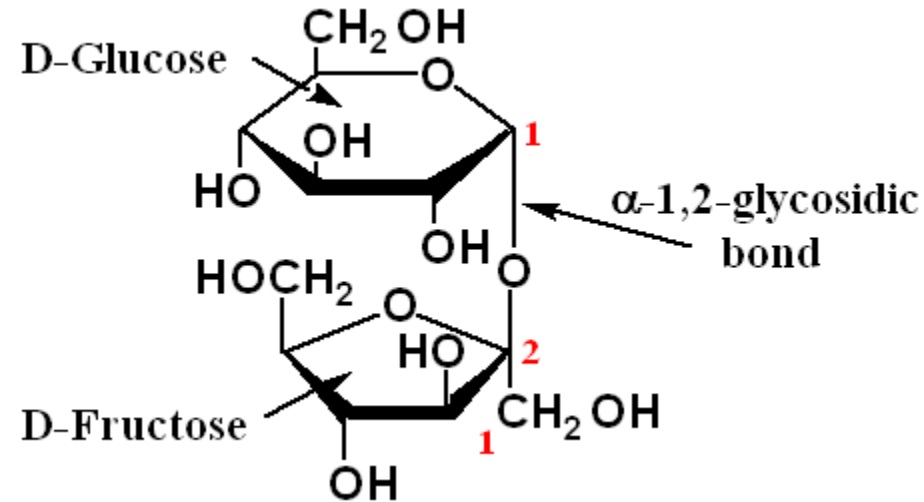


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Lactose intolerance

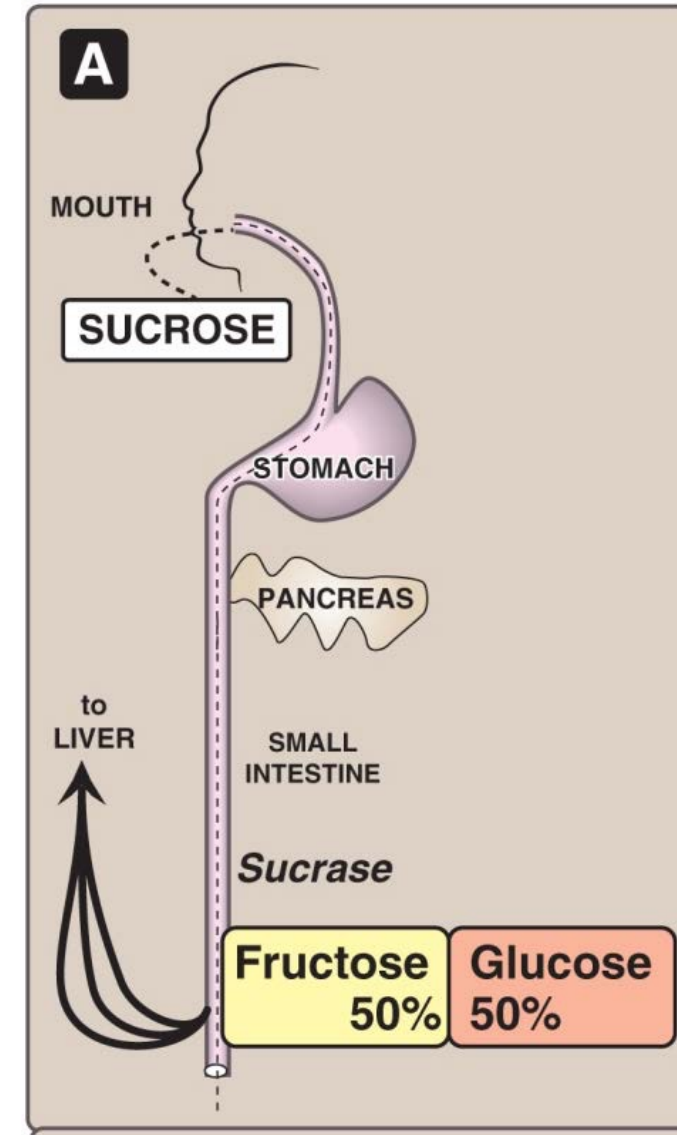


Sucrose



NON-Reducing NO FREE Ald/Keto

- Glucose + Fructose
- Glu - α -1,2 - Fru
- Non-reducing sugar: C1 and C2 not free
- Intestinal sucrase (brush border disaccharidase)
- Avoid: Hereditary fructose intolerance
- Refined sugars



High fructose corn syrup

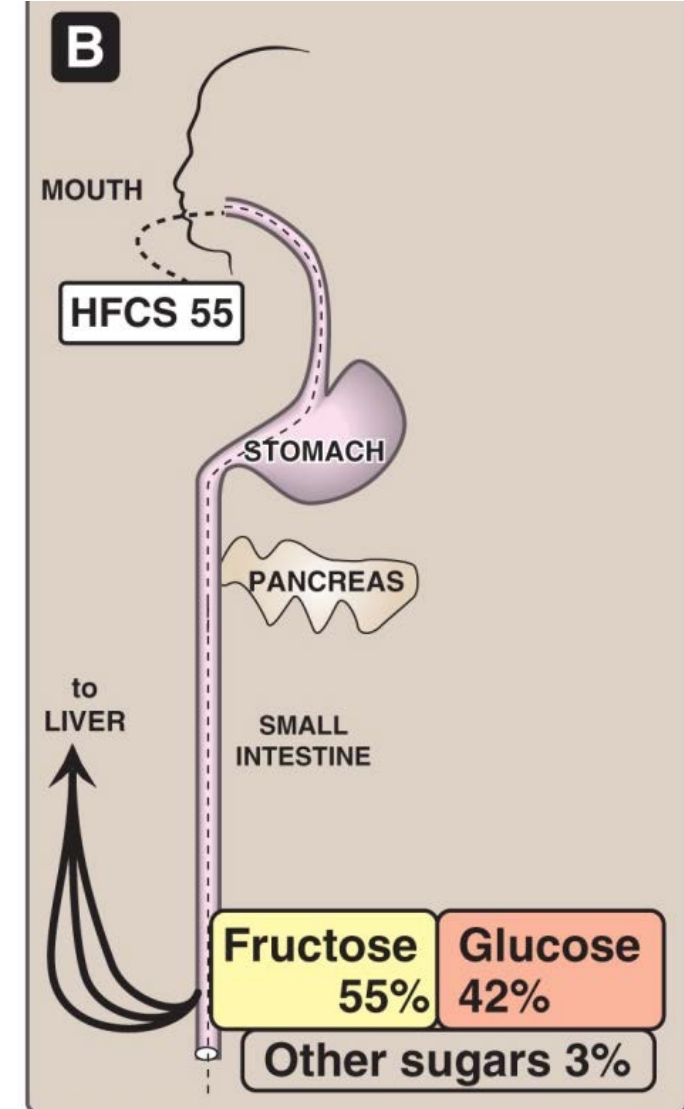
- Fructose (55%) and glucose (42%)
- Fructose: glucose ratio is greater than 1
- Mixture of monosaccharides
- Sweetener in soft drinks,

The Corny Truth About HIGH FRUCTOSE CORN SYRUP

Top 10 Foods with the Highest Quantity of HFCS:

- | | |
|---------------------|------------------------|
| 1) Yogurt | 6) Boxed Mac n Cheese |
| 2) Breads | 7) Salad Dressing |
| 3) Frozen Pizza | 8) Tomato-Based Sauces |
| 4) Cereal Bars | 9) Apple Sauce |
| 5) Cocktail Peanuts | 10) Canned Fruit |

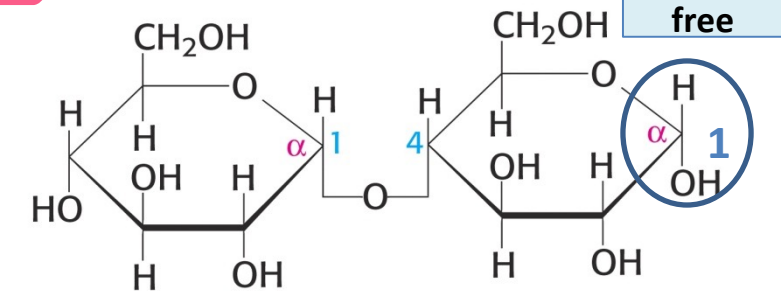
*High Fructose Corn Syrup has been linked directly to obesity, diabetes and metabolic dysfunction



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Maltose

- Two ^{C1-free} glucose units
- $\alpha 1 \rightarrow 4$ glycosidic link
- Reducing sugar
- Intermediate product of starch digestion by amylase
- Intestinal maltase (brush border disaccharidase)



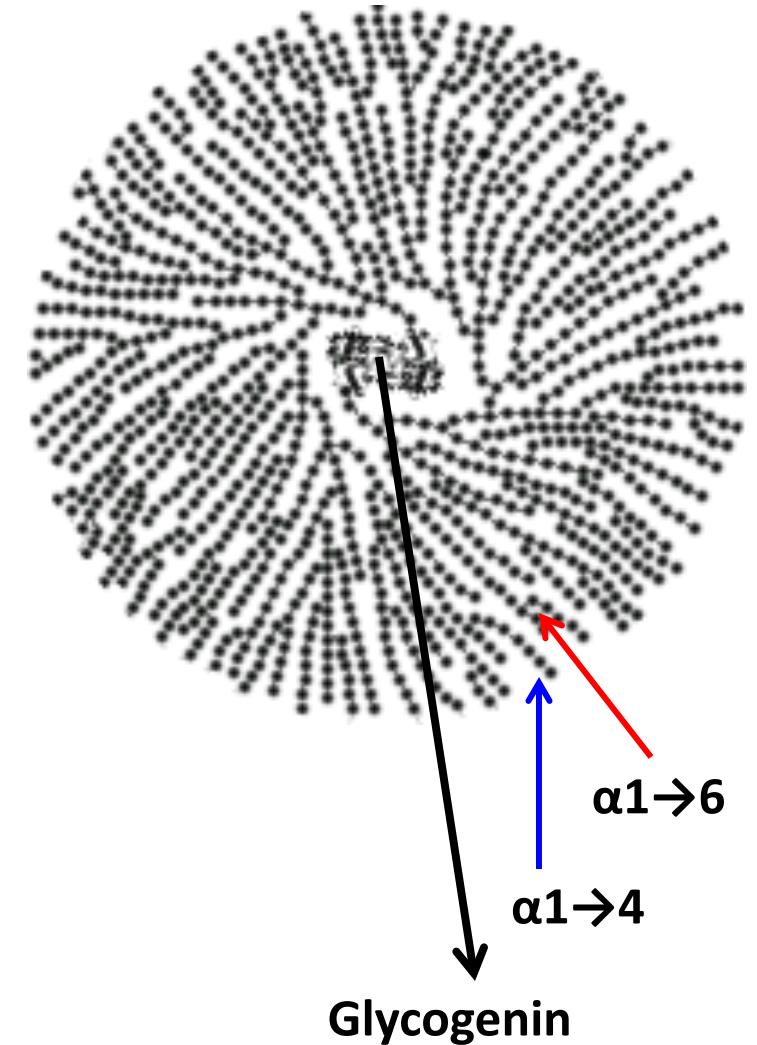
Maltose
(α -D-Glucopyranosyl-(1 $\rightarrow$ 4)- α -D-glucopyranose)

Polysaccharides

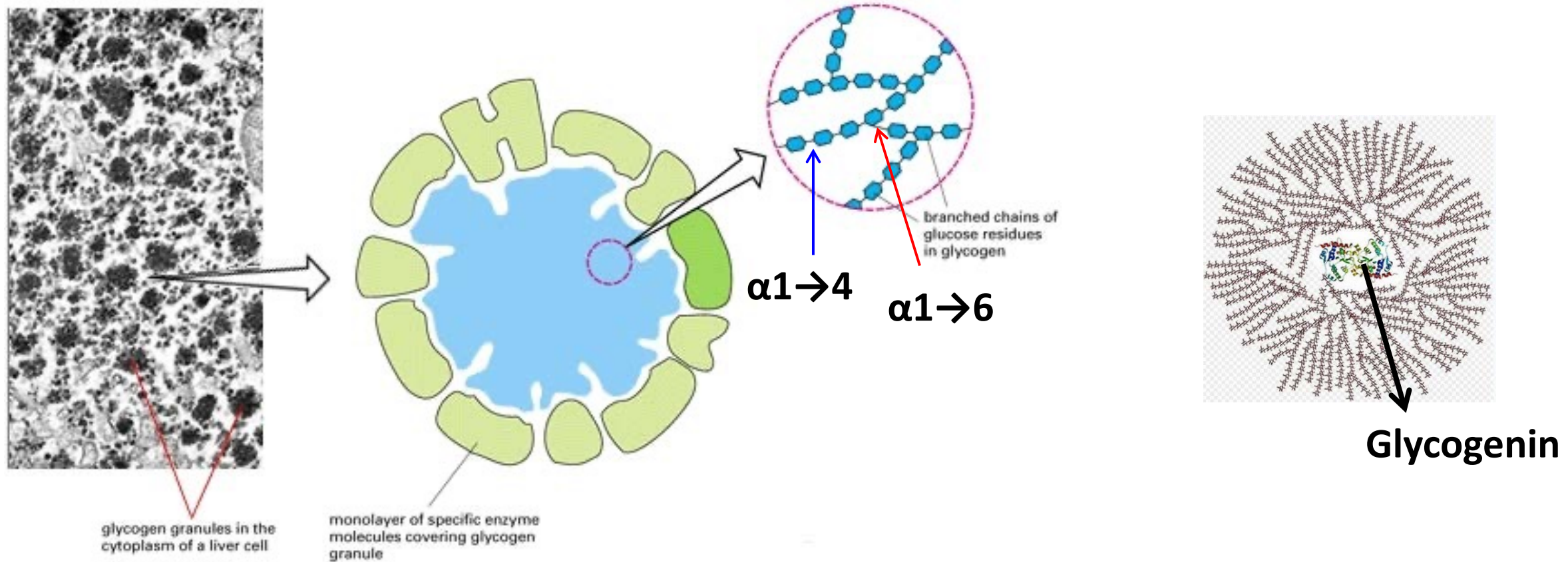
- **HOMOPolysaccharides**
 - Starch, glycogen and cellulose (glucose units)
- **HETEROPolysaccharides**
 - Glycosaminoglycans/ Mucopolysaccharides

Glycogen (storage polysaccharide)

- Liver and muscle as glycogen granules
- Glucose units
- Linear chain $\alpha 1 \rightarrow 4$ glycosidic links
- Branch point $\alpha 1 \rightarrow 6$ glycosidic links
- Branch has 8-10 units
- Glycogenin: Core protein
- Glycogen storage disorders
 - Liver or muscle
 - Abnormal structure Due to abnormal enzymes



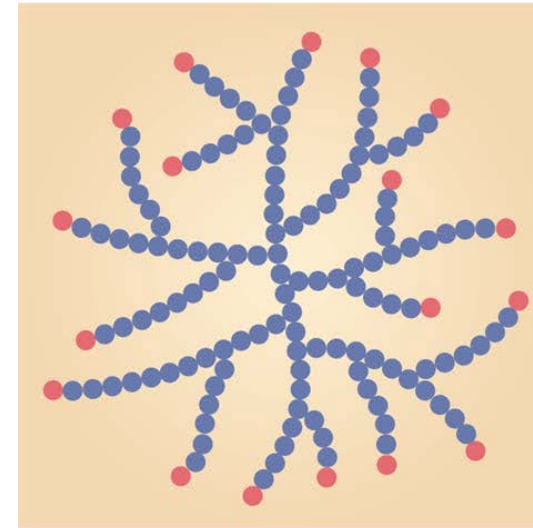
Glycogen granules in liver



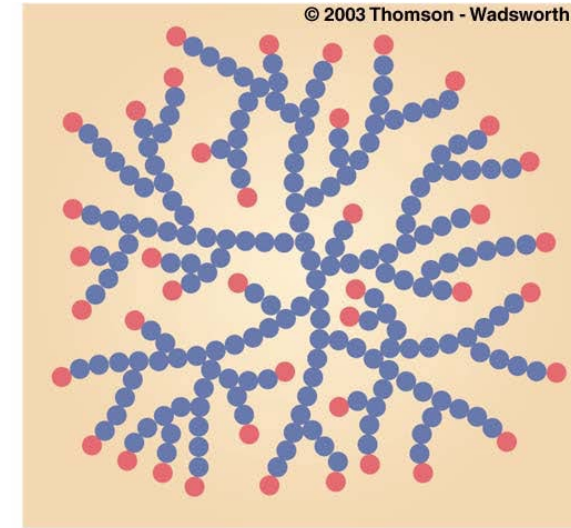
- Enzymes act on ends of branches
- Remove or add glucose

Starch (dietary polysaccharide)

- Two components:
 - Amylose = Linear unbranched
 - Amylopectin = Branched
 - Fewer longer branches
- Glucose units
- Linear chain $\alpha 1 \rightarrow 4$ glycosidic links
- Branch point: Amylopectin $\alpha 1 \rightarrow 6$ links
- Main dietary polysaccharide
- Salivary and pancreatic amylase

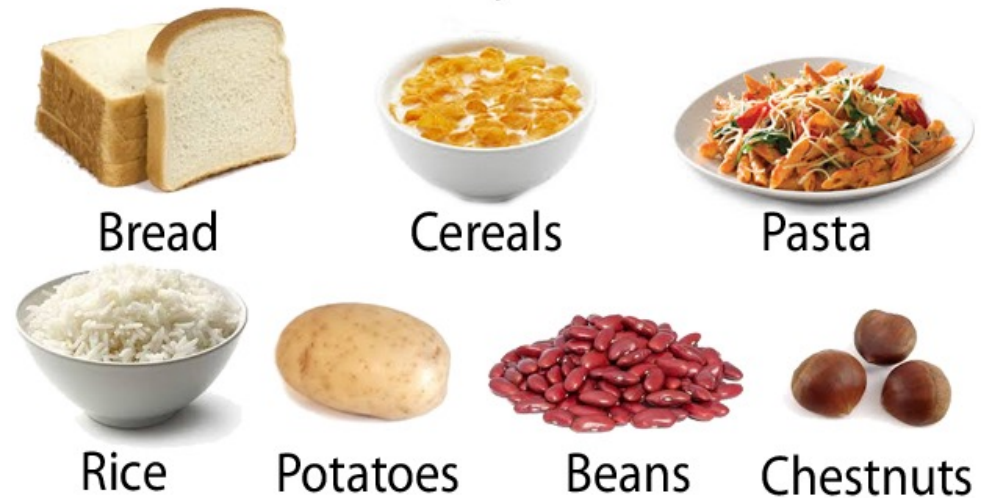


Amylopectin

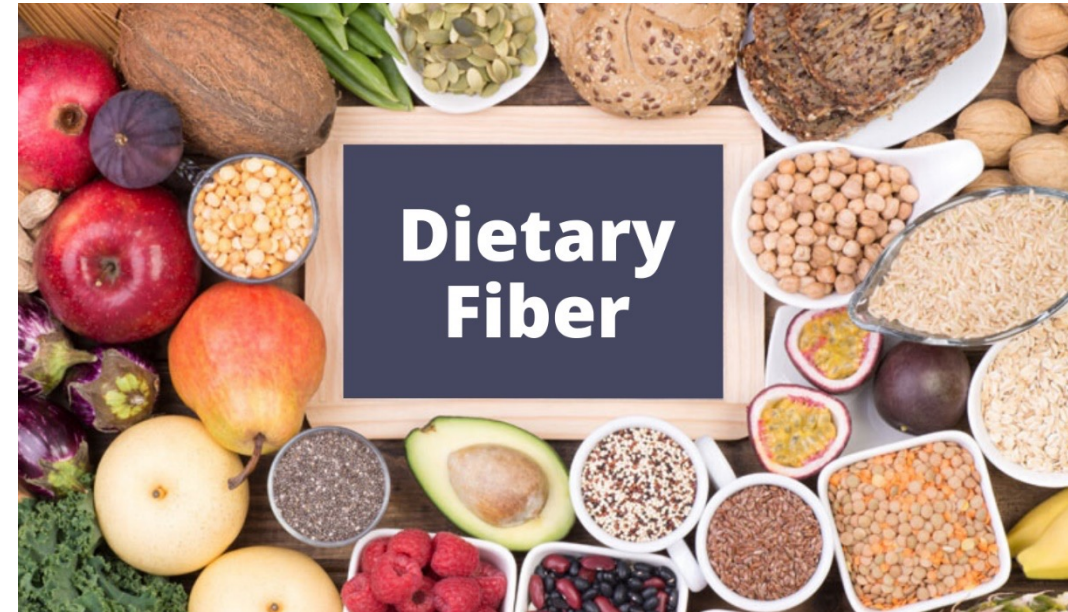
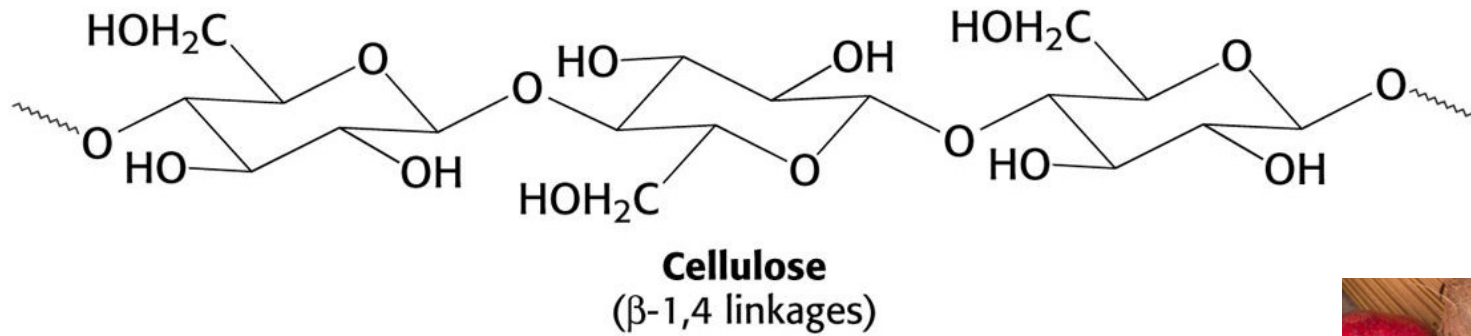


Glycogen

Starchy Foods



Cellulose (dietary fiber)



- Undigestible carbohydrate
- Unbranched homopolysaccharide
- **Glucose** units β 1 $\rightarrow$ 4 links
 - Cannot be cleaved by digestive enzymes in the human gut
- Lactase hydrolyzes β 1 $\rightarrow$ 4 links only in lactose and NOT in cellulose!!

